

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): 4K10-HS, 4K7-HS

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_0_0_0	12/27/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='4K10-HS')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='4K10-HS')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='4K10-HS')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' 'Letterbox' 'Full Height'	Value 'Native' 'Full Size' 'Custom'	Value '4:3' 'Full Width' '3D Mode'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AutoImage ¹	Value 'Normal'	Value 'Wide'	
# AutoImage example InterfaceName.Set('AutoImage', 'Normal')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'HDMI 1' 'DisplayPort'	Value 'HDMI 2' 'HDBaseT' ²	Value 'DVI-D' '3G-SDI'
# Input example InterfaceName.Set('Input', 'HDMI 1')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '1')			
Command LampMode	Value 'Constant Power' 'ECO 2'	Value 'Constant Intensity' 'Rental Mode'	Value 'ECO 1'
# LampMode example InterfaceName.Set('LampMode', 'Constant Power')			
Command MenuNavigation	Value 'Up' 'Right' 'Exit'	Value 'Down' 'Enter'	Value 'Left' 'Menu'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			

Command OnScreenDisplay	Value 'On'	Value 'Off'
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')		
Command Power	Value 'On'	Value 'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command VideoMute	Value 'On'	Value 'Off'
# VideoMute example InterfaceName.Set('VideoMute', 'On')		

¹ Normal mode supports all 4:3 input sources, whereas Wide mode supports all 16:9 input sources and most 4:3 input sources.

² HDBaseT requires an installed, optional input expansion board.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'Auto'	'Native'	'4:3'
	'Letterbox'	'Full Size'	'Full Width'
	'Full Height'	'Custom'	'3D Mode'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'HDMI 1'	'HDMI 2'	'DVI-D'
	'DisplayPort'	'HDBaseT' ¹	'3G-SDI'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			

Command LampMode	Value 'Constant Power' 'ECO 2'	Value 'Constant Intensity' 'Rental Mode'	Value 'ECO 1'
<pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LampUsage	Value Hours		
<pre># LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
<pre># OnScreenDisplay examples InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)</pre>			
Command OperationHours	Value Hours		
<pre># OperationHours examples InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)</pre>			
Command Power	Value 'On'	Value 'Off'	
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>			

¹HDBaseT requires an optional input expansion board to be installed.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

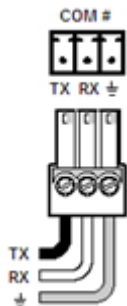
Data Bits: 8

Parity: None

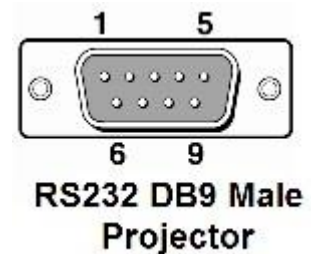
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 3D Mode	(SZP 8)
Aspect Ratio 4:3	(SZP 2)
Aspect Ratio Auto	(SZP 0)
Aspect Ratio Custom	(SZP 7)
Aspect Ratio Full Height	(SZP 6)
Aspect Ratio Full Size	(SZP 4)
Aspect Ratio Full Width	(SZP 5)
Aspect Ratio Letterbox	(SZP 3)
Aspect Ratio Native	(SZP 1)
Auto Image Normal	(AIM 0)
Auto Image Wide	(AIM 1)
Freeze Off	(FRZ 0)
Freeze On	(FRZ 1)
Input 3G-SDI	(SIN+MAIN 7)
Input DisplayPort	(SIN+MAIN 6)
Input DVI-D	(SIN+MAIN 5)
Input HDBaseT	(SIN+MAIN 8)
Input HDMI 1	(SIN+MAIN 3)
Input HDMI 2	(SIN+MAIN 4)
Keypad 0	(KEY 36)
Keypad 9	(KEY 34)
Lamp Mode Constant Intensity	(LPM 1)
Lamp Mode Constant Power	(LPM 0)
Lamp Mode ECO 1	(LPM 2)
Lamp Mode ECO 2	(LPM 3)
Lamp Mode Rental Mode	(LPM 4)
Menu Navigation Down	(KEY 42)
Menu Navigation Enter	(KEY 40)
Menu Navigation Exit	(KEY 20)
Menu Navigation Left	(KEY 39)
Menu Navigation Menu	(KEY 19)
Menu Navigation Right	(KEY 41)
Menu Navigation Up	(KEY 38)
On Screen Display Off	(OSD 0)
On Screen Display On	(OSD 1)
Power Off	(PWR 0)

Power On	(PWR 1)
Video Mute Off	(SHU 0)
Video Mute On	(SHU 1)

Appendix B. Update Commands

Aspect Ratio	(SZP?)
Freeze	(FRZ?)
Input	(SIN+MAIN?)
Lamp Mode	(LPM?)
Lamp Usage	(LIF+LSHS?)
On Screen Display	(OSD?)
Operation Hours	(LIF+LPHS?)
Power	(PWR?)
Video Mute	(SHU?)